

Real-Time Object Detection and Pose Estimation for Wiring Configurations on Qualcomm RB3 Gen2 in Industry 4.0 Applications

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Abstract—This paper describes the development and deployment of a real-time visual inspection system for identifying daisy chain wiring configurations and estimating their pose using a two-stage AI approach: YOLO (You Only Look Once) for object detection and HRNet (High-Resolution Network) for keypoint-based pose estimation [5]. The system runs on the Qualcomm RB3 Gen2 Vision Kit (QCS6490), an edge-AI platform optimised for high-performance, low-latency embedded processing. By exploiting the AI acceleration capabilities of the RB3 hardware, the system achieves rapid inference while maintaining high detection and pose accuracy. Designed to operate entirely on-device, the solution removes cloud dependencies, thereby reducing latency, improving security, and increasing operational reliability. This work demonstrates strong potential for integration into Industry 4.0 environments, enabling automated, in-line quality control in smart manufacturing facilities.

I. INTRODUCTION

In manufacturing—particularly automotive and electronics—correct assembly of wiring harnesses such as daisy chains is essential for product reliability and safety [1]. Manual inspection remains widespread but suffers from low scalability, inconsistency, and human error. Deep learning-based computer vision, especially the YOLO family of algorithms [2], offers real-time, accurate object detection. For industrial deployment, edge inference is required to meet latency, reliability, and security demands, avoiding cloud reliance.

This study presents an embedded AI system that performs object detection and pose estimation for daisy chain wiring assemblies directly on the Qualcomm RB3 Gen2 Vision Kit (QCS6490) [3]. While YOLO is used for detection, pose estimation is implemented using the High-Resolution Network (HRNet) architecture [5], known for preserving high-resolution representations throughout the network to achieve superior keypoint localization accuracy. The system supports both live video streams and static snapshots, offering a compact, energy-efficient, and high-performance solution for automated inspection.

II. METHODS

A. Hardware Platform

The deployment target was the Qualcomm RB3 Gen2 Vision Kit (QCS6490), which integrates a Hexagon DSP, an Adreno GPU, and a dedicated AI Engine. This combination delivers efficient on-device AI inference while maintaining low power consumption—making it well-suited to continuous industrial operation [3].

B. Software and Libraries

To implement cascaded inference for real-time video analysis, we utilized the `gst-ai-daisychain-detection-pose` pipeline, a GStreamer-based architecture optimized for Qualcomm platforms [4]. This pipeline facilitates sequential object detection followed by pose estimation on the detected regions, supporting input sources such as live camera feeds, pre-recorded video files, or RTSP streams.

The pipeline begins with an input source that is branched using a tee element to enable parallel processing. The first branch feeds into the initial inference bin, which performs object detection. This sub-pipeline includes the `qtmvconverter` for pre-processing and format conversion, an inference module for running the YOLO object detection model, and a post-processing stage to extract relevant metadata. The detection results are then passed to a `metamuxer`, which embeds the metadata into the GStreamer buffer for downstream modules.

The second inference bin processes the original input again, utilizing the detection metadata from the first inference to perform pose estimation specifically on the regions of interest. In this stage, an HRNet-based quantised pose estimation model is deployed for accurate keypoint localization while maintaining low-latency performance. Similar to the first inference path, it comprises a `qtmvconverter`, inference engine, post-processing block, and `metamuxer` for output formatting.

The outputs of the two inference paths are synchronized and passed through a second tee, which routes the processed data to a set of Qualcomm-specific visualization elements. These include `qtvsplit` for parsing metadata streams, `qtvcomposer` for merging pose and detection metadata, and `qtioverlay` for rendering visual annotations such as bounding boxes and skeletal poses. Finally, the annotated video stream is directed to both `waylandsink` for real-time display and `filesink` for persistent storage.

This modular and parallelized approach allows for efficient, low-latency cascaded inference and visualization.

C. Model Training and Deployment

A YOLOv8 object detection model [2] was trained on a custom dataset of daisy chain wiring assemblies captured under varied lighting, background, and orientation conditions. Data augmentation—including rotation, scaling, and brightness adjustments—was applied to improve robustness against production floor variations.

For pose estimation, an HRNet (High-Resolution Network) model [5] was trained on a keypoint-labelled subset of the same dataset. This ensured both models were optimised for the exact geometry

and connector configurations present in the target environment.

Following training, both YOLOv8 and HRNet models were quantised to INT8 using the Qualcomm Neural Processing SDK [3]. Quantisation reduced model size and compute requirements, enabling efficient deployment without significant loss of accuracy. Graph-level optimisations were applied to fully utilise the RB3's DSP and GPU acceleration, ensuring low-latency operation suitable for high-speed production lines.

The quantised models were integrated into Qualcomm's gst-ai-daisychain-detection-pose pipeline for on-device inference. Live video was captured via qtqmmfsrc and split into two processing paths. In the detection path, qtimlvconverter handled preprocessing (color conversion, scaling, normalization) before qtimlflite executed the YOLOv8 model. Detection results were processed by qtimlvdetection, applying thresholds and generating bounding boxes for regions of interest.

In the pose estimation path, detection metadata was used to crop regions via qtvsplit before running the quantised HRNet model in qtimlvpse. This module generated skeletal keypoints and applied orientation calculations for quality verification.

The outputs of both inference paths were synchronised and merged via qtvcomposer, annotated using qtioverlay, and rendered locally through waylandsink. This manual training and optimisation process allowed both models to achieve real-time inference—averaging 27.3 ms per frame for YOLOv8 (~36.6 FPS) and 31.5 ms per frame for HRNet (~31.7 FPS)—while running entirely on-device without cloud dependencies.

The HRNet pose estimation model was optimised in a similar fashion, with quantisation and DSP/GPU acceleration ensuring real-time operation.

D. Pose Estimation

Following detection, pose was estimated using HRNet keypoint outputs [5]. Keypoints were determined relative to connector edges, allowing calculation of angular alignment for quality verification.

III. RESULTS

The deployed system achieved real-time performance on the RB3 Vision Kit.

- YOLO Object Detection and Pose Estimation: Average inference time 27.3 ms per frame (~36.6 FPS)
- Pose Estimation (HRNet): Average inference time 31.5 ms per frame (~31.7 FPS)

Additional HRNet runtime performance metrics from live testing:

- Average FPS: 30.0
- Current FPS: 29.9
- Frames Rendered: 2991
- Frames Dropped: 9

Both live-stream and snapshot inspection modes performed

reliably. In live-stream mode, detection was stable and smooth, suitable for high-speed production lines. In snapshot mode, the system provided instantaneous, accurate feedback—ideal for targeted inspections.



Figure 1 Yolo 10 Object detection with pose estimation output

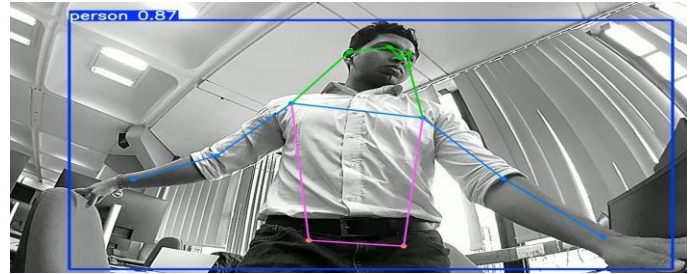


Figure 2 yolo pose estimation output

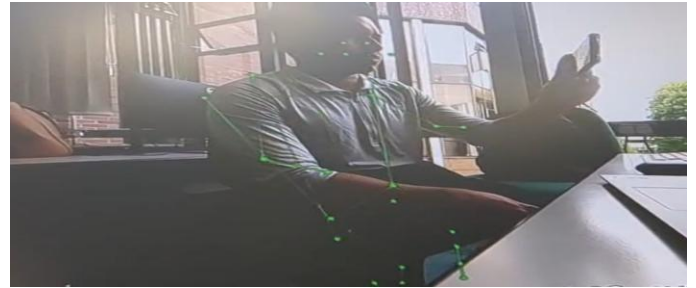


Figure 3 HRNet Quantised Pose Estimation Output

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